Junchao Chen

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EDUCATION

Anhui University
B.S. in Economic Statistics | GPA: 88.91/100 Sep 2024 –

RESEARCH EXPERIENCE

Probability Limit Theorems for Dependent Variables and Their Applications to EV Regression Models 2026 – 2027
Provincial research grant | Project leader

- **Limit theory:** Studied complete convergence and complete f-moment convergence for maximal randomly weighted sums of conditionally rowwise m-WOD arrays, including logarithmic endpoint cases.
- **Sharpness analysis:** Examined sign-block constructions to identify thresholds for block growth and logarithmic moments, supported by examples and counterexamples.
- **EV regression:** Applied the convergence results to a stabilized repeated-measurement errors-in-variables estimator and analyzed conditions for complete consistency.

LLM–SLM Collaboration for Sarcasm Detection
Ongoing research

- Exploring collaborative frameworks between large and small language models for sarcasm detection.
- Studying model effectiveness, efficiency, and complementary reasoning capabilities.

Wiener Degradation Models with Nonlinear Drift and Stochastic Volatility for Remaining Useful Life Prediction
Reliability statistics research

- **Degradation modeling:** Studied nonlinear-drift Wiener processes with stochastic volatility for threshold-based remaining useful life (RUL) prediction.
- **Functional estimation:** Analyzed local-polynomial mean estimation, bipower variation, and FPCA-based recovery of variance trajectories from repeated degradation paths.
- **Reliability inference:** Distinguished conditional variance clocks from marginal variability and examined admissibility conditions for cumulative variance estimates.
- **Simulation study:** Evaluated 300 benchmark and 900 controlled datasets to study sampling frequency, drift contamination, and latent-volatility effects on prediction.
- **Model validation:** Identified cases where low curve-estimation error coexists with invalid variance clocks or threshold-crossing probabilities.

AWARDS AND COMPETITIONS

National Encouragement Scholarship 2025–2026
Academic Excellence Scholarship 2024–2025
M Award, Mathematical Contest in Modeling 2025–2026
Provincial Second Prize, National Undergraduate Statistical Modeling Competition 2025–2026

SKILLS

Languages: CET-6: 575

Technical Skills: R / Python / PyTorch / LaTeX / functional data analysis / stochastic degradation modeling / dependent-data asymptotics

Independent Study: Stanford CS229, CS230, CS224N, and CS231N course materials (machine learning, deep learning, natural language processing, and computer vision).